

Requirements Exercise

The company XYZ has signed a new contract with the library ABC to develop a new library system. To reduce the cost of the development, the XYZ has decided to outsource the implementation to India. XYZ develops the requirements specification, the company in India implements the system. XYZ has written a first version of the requirements document. In the following we present the requirements document for the library system; for simplicity, non-functional requirements are omitted.

This document describes a system for a library.

The main functionalities of the system allow

- borrowing and returning books in a library;
- getting the list of books by a particular author or in a particular subject area,
- finding out the list of books currently checked out by a particular borrower,
- finding out what borrower last checked out a particular copy of a book.

Template to create Requirements doc

- Introduction of the system
- Revision history of the document
- Abbreviations, Definitions, Acronyms, Glossary of terms
- Basic Structure of Requirement (next slide)
 - Functional Requirements
 - Non-Functional Requirements (optional at this stage)
 - Goals of the Implementation (optional at this stage)

Template of writing Requirements

ID	Defines a unique identifier of the requirement. The requirements are ordered by topic.
Title	Title of the requirement
Description	A description of the requirement.
Priority	Defines the order in which requirements should be implemented. Priorities are 1, 2, 3, and 4 (highest to lowest). Requirements of priority 1 are mandatory for the first implementation; requirements of priority 2 are mandatory for the final implementation; priority 3 is used for features that are optional but the client would like to have it; priority 4 is used for optional features.
Risk	Specifies (1) the risk of not implementing the requirement, and (2) a probability that this feature is not implemented. The first one shows how critical the requirement is to the system as a whole; the second one, the probability, is a percentage 0...100.

Template of writing Requirements

The following risk levels are defined over the impact of not being implemented correctly.

- Critical: it will break the main functionality of the system. The system cannot be used if this requirement is not implemented.
- High: it will impact the main functionality of the system. Some functions of the system could be inaccessible, but the system can generally be used.
- Medium: it will impact some system features, but not the main functionality. The system can still be used with some limitations.
- Low: the system can be used without limitations, but with some workarounds.

Acronyms, Glossary etc.

The following table explains the terms and abbreviations used in the document:

Term/abbreviation	Explanation
LS	Library System
BB	Borrow Book

Word	Explanation
User	Any person who uses the system and is registered within the system. It means that he or she has a user login.

ID R1.01

Title: Books

Description: The system stores books. The status of a book is either available or borrowed.

Priority: 3 **Risk:** Low / 50%

ID R1.02

Title: Books \ book copies

Description: The system also stores book copies. A book can have several copies. The status of a book copy is either available or borrowed.

Priority: 3: **Risk:** Low / 35%

ID R1.03

Title: Books \ book copies \ copies of a book

Description: The system must be able to show the amount of copies of a particular book. This functionality is only available to Staff users.

Priority: 3 **Risk:** Low / 30%

ID R1.04

Title: Books \ repair

Description: The system shall provide functionality to repair book copies. The status of a book copy is either available or borrowed or broken. If the book is in status broken, this functionality repairs the book by setting a string attribute to available. This functionality is available to any user.

Priority: 3 **Risk:** High / 10%

ID R1.05

Title: Books \ borrow

Description: The system shall provide functionality to borrow book copies. The user selects an available book and borrows it. This functionality is only available to Borrower users.

Priority: 3: **Risk:** Critical / 10%

ID R1.06

Title: Books \ return

Description: The system shall provide functionality to return book copies. The user returns a book if the book status is borrowed. This functionality is available to any user.

Priority: 3 **Risk:** Low / 30%

ID R2.01

Title: Users \ User Roles

Description: There are two kind of users: Administrator and Borrower. The Administrator user can access any functionality, and the Borrower user only the functionalities defined in R1.01, R1.02, and R1.03.

Priority: 2 **Risk:** High / 10%

ID R2.02

Title: Library

Description: Any Administrator user can check how many books are available and how many books are borrowed. Furthermore, the Administrator user can check how many books are broken. To borrow, or return a book, the user has to log in first.

Priority: 3 **Risk:** Low / 40%

ID R3.01

Title: List of books \ By user

Description: The system shall be able to display a list of books authored or co-authored by a given author. The list shall be ordered in chronological order. If the author published more than one book in the same year, the list should be also order by the title.

Priority: 1 **Risk:** Low / 70%

ID R3.02

Title: List of books \ By topic

Description: The system shall be able to display a list of books in given subject area. The list shall be ordered by topic and chronological order. If there is an author who published more than one book in the same year, and area, the list should be also order by the title.

Priority: 1 **Risk:** Low / 70%

ID R3.03

Title: List of books \ Checked out by user

Description: The system shall be able to display a list of books that a given user has checked out. The list should be ordered by date.

Priority: 1 **Risk:** Low / 70%

ID R3.04

Title: List of books \ Checked out by book

Description: The system shall be able to finding out what borrower last checked out a particular copy of a book.

Priority: 1 **Risk:** Low / 70%

Do the following

- 1 Define the stakeholders of the system.
- 2 Given the above requirements document, give examples (extracted from this document) of quality goals that are not satisfied.

Solution 1: Stakeholders of the Library System

Vendor XYZ
Library ABC
Developers in India
Users

Solution 2:

On review of Req. Doc., following quality goals were not satisfied:

- **not Consistent:** Requirement R2.0 says: the "Borrower" user only to the functionalities defined in R1.01, R1.02, and R1.03". And requirement R1.03 says "This functionality is available to staff users".
- **not Consistent:** Requirement R1.01 says: "The status of a book is either "available" or "borrowed" ". Requirement R1.04 says "The status of a book are either "available" or "borrowed" or "broken".
- **not Unambiguous:** "Book copy" is never described (missing in the glossary), it is not clear if this is a physical copy, or a photocopy, or something else.
- **not Abstract:** Requirement R1.04 says "this functionality repairs the book setting a string attribute to "available".

On review of Req. Doc., following quality goals were not satisfied:

- **not Prioritized:** The give priorities are not correct. The important features have low priorities, and the less important features have high priorities.
- **not Complete:** A functionality "log in" for users is missing (and a lot more...)
- **not Delimited:** R2.02 talks about two separate concerns, (1) Administrator checking on status, and (2) that the user has to be logged in to borrow or return a book.
- **not Traceable:** The revision history is missing (tracing of who wrote what in the req. document).

Prepare Requirements Doc for the software for automating the activities of a 5-star hotel.

Hotel Automation Software: A hotel has a certain number of rooms. Each room can be either single bed or double bed type and may be AC or Non-AC type. The rooms have different rates depending on whether they are of single or double, AC or Non-AC types. The room tariff however may vary during different parts of the year depending upon the occupancy rate. For this, the computer should be able to display the average occupancy rate for a given month so that the manager can revise the room tariff either upwards or downwards by a certain percentage. Guests can reserve rooms in advance or can reserve rooms on the spot depending upon availability of rooms. The receptionist would enter data pertaining to guests such as their arrival time, advance paid, approximate duration of stay, and the type of the room required.

Depending on this data and subject to the availability of a suitable room, the computer would allot a room number to the guest and assign a unique token number to each guest. If the guest cannot be accommodated, the computer generates an apology message. The hotel catering services manager would input the quantity and type of food items as and when consumed by the guest, the token number of the guest, and the corresponding date and time. When a customer prepares to check-out, the hotel automation software should generate the entire bill for the customer and also print the balance amount payable by him. Frequent guests should be issued identity numbers which would help them to get special discounts on their bills.

Practice Examples

- Withdraw cash from ATM
- Withdraw cardless cash from ATM
- Ordering food online
- Online ticket booking for GSRTC
- Library Book Search

Smart Campus Event Management System

A university intends to deploy a centralized platform to manage academic, cultural, and technical events conducted by various departments and student clubs. Event organizers should be able to propose events with details such as title, description, venue, capacity, schedule, and required approvals. The system must route proposals through an approval workflow involving faculty coordinators and the student affairs office. Once approved, events are published on a campus calendar accessible to all students and staff. Students can browse events by category, register for participation, and receive confirmation along with QR-code-based entry passes. Organizers must be able to view registration statistics and download participant lists before the event.

On the operational side, the system should support on-site check-in using QR scanning to mark attendance and prevent duplicate entries. Automated notifications must be sent for schedule changes, venue updates, and event reminders. Role-based access control is required for administrators, faculty coordinators, organizers, and students. The system should integrate with the university's email service and optionally with the student information system to verify enrollment status. Reliability during peak registration periods, audit logs for approvals, and data retention for at least five academic years are important non-functional requirements.

- User Management

- FR-1: The system shall allow administrators to create, modify, deactivate, and assign roles to user accounts.
- FR-2: The system shall authenticate users using institutional login or system credentials.
- FR-3: The system shall authorize access based on assigned roles.

- Event Proposal and Approval

- FR-4: The system shall allow organizers to submit event proposals with title, description, venue, date, time, category, capacity, and resource requirements.
- FR-5: The system shall route proposals to designated faculty coordinators for approval.
- FR-6: The system shall support approval, rejection, or request-for-modification actions with comments.
- FR-7: The system shall notify organizers of approval decisions.

● Event Publication and Discovery

- FR-8: The system shall publish approved events on a campus-wide calendar.
- FR-9: The system shall allow users to filter events by category, date, organizer, and venue.
- FR-10: The system shall display real-time seat availability.

● Registration and Ticketing

- FR-11: The system shall allow students to register for events until capacity is reached.
- FR-12: The system shall generate unique QR codes for each registered participant.
- FR-13: The system shall send registration confirmations via email.

- Attendance Management

- FR-14: The system shall allow organizers to scan QR codes for attendance verification.
- FR-15: The system shall prevent duplicate check-ins.
- FR-16: The system shall timestamp attendance records.

- Notifications and Updates

- FR-17: The system shall notify participants of event changes or cancellations.
- FR-18: The system shall send reminder notifications before events.

- Reporting and Analytics

- FR-19: The system shall provide organizers with participant lists and attendance statistics.
- FR-20: The system shall provide administrators with historical event and participation reports.
- FR-21: The system shall support data export in standard formats.

- Audit and Logging

- FR-22: The system shall log all approval decisions and administrative actions.
- FR-23: The system shall maintain logs for compliance audits.

Pharmacy Inventory & Prescription System

A neighborhood pharmacy wants to digitize its medicine inventory and prescription handling process. Pharmacists should be able to register new medicines with batch numbers, expiry dates, suppliers, pricing, and storage requirements. The system must track stock levels in real time and automatically flag low-stock and near-expiry items. Doctors' prescriptions may be entered manually or uploaded as scanned images, and pharmacists should record which medicines are dispensed against each prescription. Customers may be registered with basic demographic details and purchase history for regulatory and loyalty purposes.

Pharmacy Inventory & Prescription System ...cont'd

From a compliance and safety perspective, the system must enforce rules for controlled substances, including identity verification and mandatory record keeping. Management should be able to generate daily sales reports, supplier-wise purchase summaries, and regulatory compliance logs. Integration with billing and GST invoice generation is required. The system must support multiple counters operating simultaneously without inventory inconsistencies. Data accuracy, transaction atomicity, backup, and recovery are critical, and access should be restricted so that only licensed pharmacists can modify prescription and drug records.

Public Transport F/b & Service Analytics Platform

A municipal transport authority plans to collect structured passenger feedback across buses, metro lines, and suburban trains. Passengers should be able to submit feedback through a mobile-friendly web interface by selecting route, vehicle number, time of travel, and issue category such as cleanliness, punctuality, staff behavior, or overcrowding. Optional photo uploads and geolocation tagging should be supported. Anonymous submissions are allowed, but registered users may track the status of their complaints. Each feedback item must be automatically routed to the appropriate depot or service division.

Public Transport F/b & Service Analytics Platform

... cont'd

For administrators, the platform must provide dashboards showing trends by route, depot, time of day, and complaint category. Supervisors should be able to assign cases to inspectors, record actions taken, and close complaints with resolution notes. Escalation rules should trigger if complaints remain unresolved beyond specified service-level thresholds. The system should support data export for policy analysis and long-term planning. Performance during peak commuting hours, protection of personal data, and mechanisms to prevent spam or abusive submissions are essential quality attributes.